🍔 MEALDASH

Delivery Operations Analytics

Diagnosing why delivery times vary across 45,000+ quick-commerce orders and turning it into three specific, actionable recommendations.

Dataflow Gen2 | SQL | Microsoft Fabric | Power BI

Prepared for: Analytics Leadership Review

THE BUSINESS PROBLEM

Nobody knew why delivery times varied or how to find out fast

MealDash's delivery times swing widely: some orders arrive in 10 minutes, others take almost an hour. Diagnosing a single slow week meant a data analyst manually digging through raw records for a day or two, with no standing answer to the question ops kept asking:

> ***"Why was it slow and what can we actually do about it?"***

This project answers 10 specific business questions — moving from the overall scale of the problem, to the levers ops can control (traffic, distance, staffing, bundling) versus the ones they can't (weather), and finally to whether performance is trending better or worse over time.

10 – 54 min

Delivery time range observed

1–2 Days

To manually diagnose a slow week — before this project

Minutes

To answer the same question — after this project

DATASET & SCOPE

A real, messy, 45,000-order dataset nothing synthetic

What's in the data

- Delivery timestamps & GPS coordinates
- Weather conditions & road traffic density
- Delivery agent ratings & vehicle type
- Festival-day flags
- ≈45,000 real orders — Zomato Delivery
- Operations dataset (Kaggle)

✓ In Scope

- Traffic
- Weather
- Distance
- Vehicle type
- Agent performance
- Order bundling
- Festival effects

✗ Out of Scope

- Pricing
- Restaurant menu data
- Customer app experience

Left out on purpose — pulling them in would dilute a focused delivery-time analysis.

Note: no "promised delivery time" field exists, so SLA breach rate is benchmarked against segment averages instead of a formal target.

APPROACH

A governed, repeatable pipeline not a one-off manual pull

1

Version on GitHub

Raw CSV pushed to a refreshable source, not a static upload

2

Clean in Dataflow Gen2

Power Query: nulls, formats, categories + Haversine Distance_km column

3

Load to Lakehouse

Dataflow Gen2 loads cleaned data into a governed Lakehouse

4

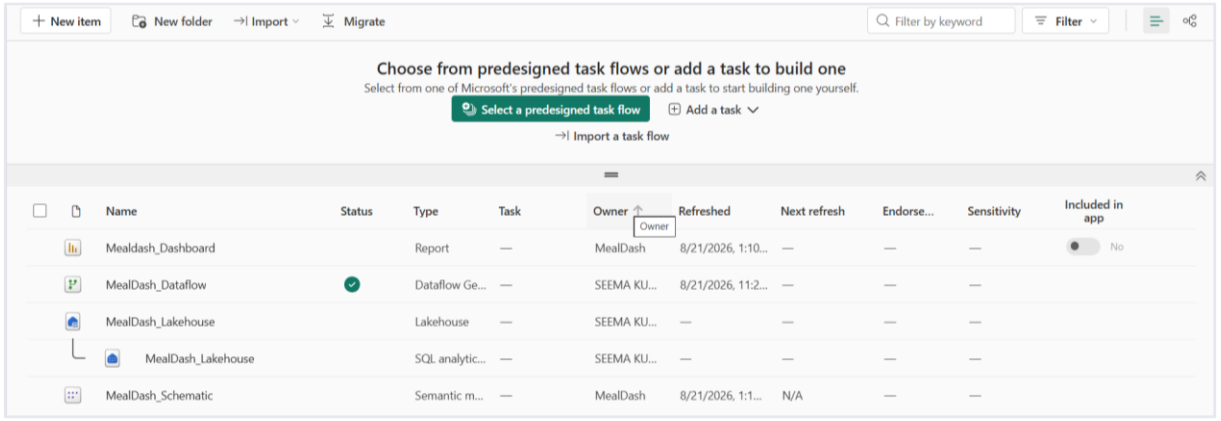
Query in SQL

10 business questions answered via the SQL Analytics Endpoint

5

Visualize in Power BI

3-page live dashboard on the same semantic model



Fabric workspace: Lakehouse, Dataflow Gen2 & SQL Analytics Endpoint under one governed roof

TOOLS & TECHNOLOGIES

Each tool earns its place in the pipeline

1 Dataflow Gen2 · Power Query

Pulls the raw CSV from GitHub, then cleans it: nulls, formats, categorical standardization, and a Haversine Distance_km column.

2 SQL · Fabric Analytics Endpoint

Answered all 10 business questions using aggregation, CASE-based bucketing, and a window function (7-day rolling average).

3 Microsoft Fabric · Lakehouse

A repeatable, governed pipeline: the cleaned data lands in a Lakehouse table instead of a one-off manual upload.

4 Power BI

A 3-page live, interactive dashboard with DAX measures, built on the same governed semantic model as the SQL layer.

KEY INSIGHTS · 1 OF 3

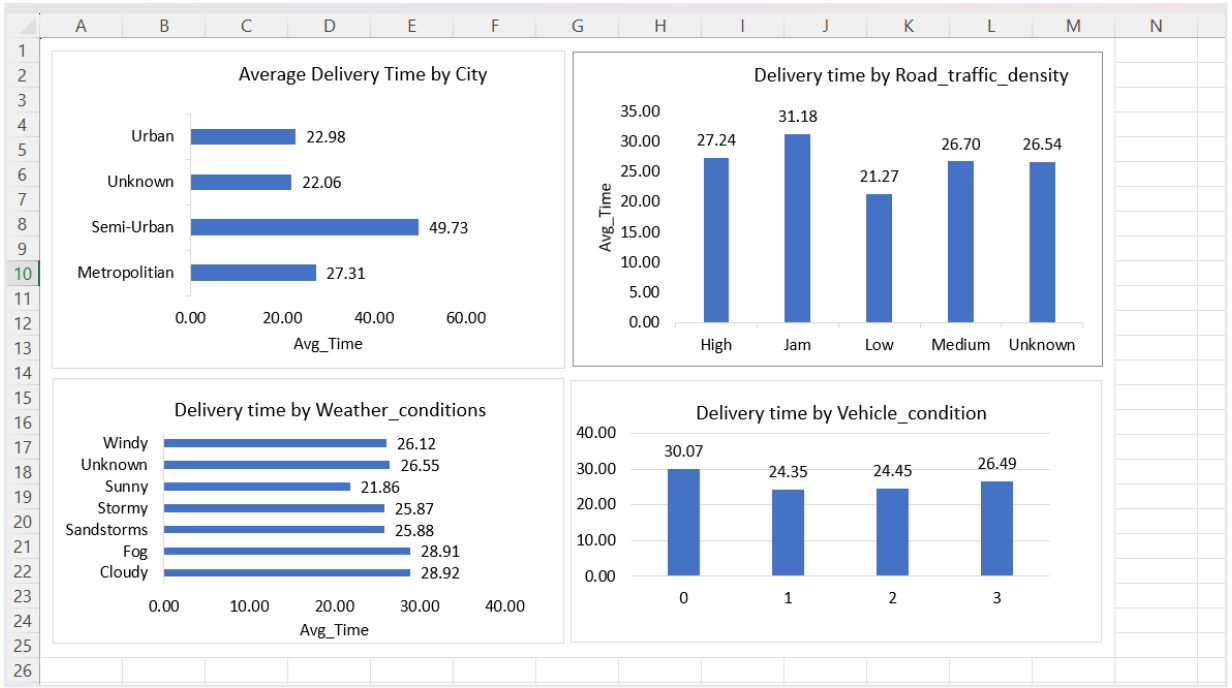
Baseline, city, traffic & weather

26 min
Overall avg (10–54 min range)

27 min
Metropolitan — slowest by volume (34,000+ orders)

+10 min
Jam traffic vs. Low traffic

28 min
Fog & Cloudy — slowest weather (not storms)



Distance, vehicle, agent rating & bundling



Distance

Crossing the 5 km mark is the real cost — time plateaus after that, it isn't proportional to distance.

Vehicle type

Minor effect (~3 min) between fastest and slowest vehicle.

Agent rating

4.5+ rated agents deliver 10–13 min faster than every lower bucket — the clearest pattern in the dataset.

Bundled orders

2–3 bundled orders more than doubles delivery time (22 → 47 min), and the cost accelerates with each extra stop.

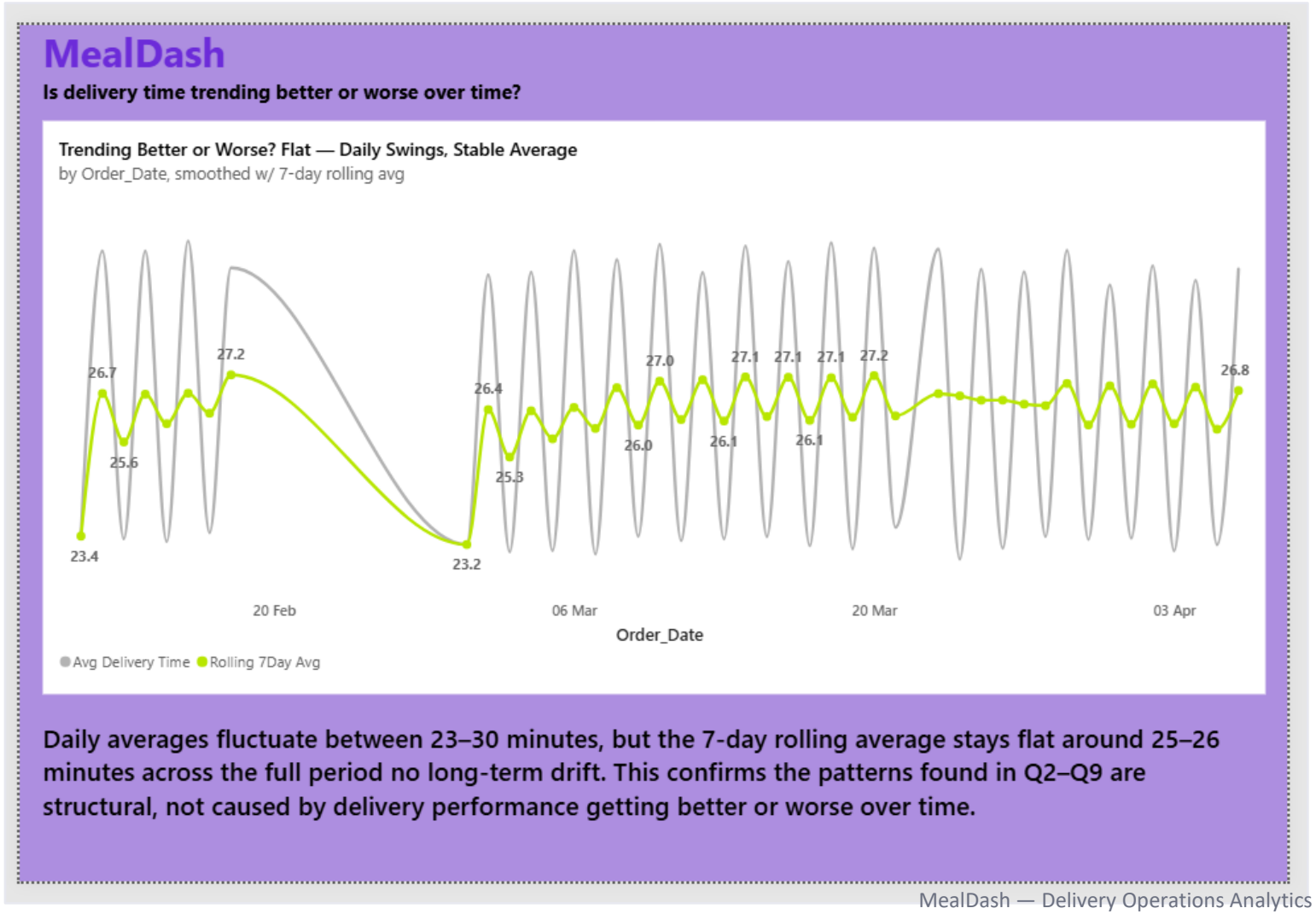
Festival impact & the trend over time

75–80%

Festival days are slower than normal days — the largest single effect found across all 10 questions

Flat

7-day rolling average holds steady around 25–26 min — no long-term drift; patterns are structural



LIVE DASHBOARD

A 3-page Power BI dashboard, built for self-serve exploration



Page 1 — Executive Overview

Total orders, avg delivery time, festival impact %, and slow-delivery rate at a glance.

Page 2 — Delivery Drivers Deep Dive

Distance, rating, vehicle condition and bundling, all slicer-driven.

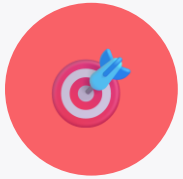
Page 3 — Trend Over Time

Daily averages smoothed with a 7-day rolling average to separate noise from signal.

🔗 Open the Live Interactive Dashboard

RECOMMENDATIONS

Three findings ops can act on this week



Pre-plan festival-day staffing

Festival deliveries take 75–80% longer — the single largest effect found. Low volume (896 orders) but high impact makes this a high-ROI, low-effort fix: targeted staffing on known festival dates, not a company-wide change.



Cap bundled deliveries at 1 extra order

2–3 bundled orders more than doubles delivery time (22 → 47 min), and the cost accelerates non-linearly. Capping at 1 extra order for time-sensitive deliveries captures most of the efficiency without the steep penalty.



Use agent ratings for order routing

4.5+ rated agents deliver in 24 min vs. 34–37 min for every lower bucket — a 10+ minute gap. A strong, easy-to-access signal for prioritizing time-sensitive orders, even though causal direction isn't fully resolved.

Traffic and weather matter too — but they're conditions ops can't directly control. These three are things ops can act on now.

WHAT'S NEXT

Future Work

1. Validate the festival-day effect against a full year of festival dates
2. Replace straight-line (Haversine) distance with actual road-route distance
3. Investigate the direction of causation between agent rating and delivery speed
4. Add a live data connection to support ongoing monitoring, not one-time diagnosis

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Thank you

Open to opportunities, collaborations, and conversations around data analytics.

github.com/seema-kri/MealDash